

Assignment 3

Information Theory for Data Science

328158, Maleki, Amir
328728, Masteri Farahani, Farzaneh
323337, Akhane, Mohammad Amin
327167, Noshiravanbaboli, Yasaman

December 27, 2023

Data Science and Engineering

Politecnico di Torino

2023-2024

1 Distortion

Considering the random variable X with the following probability density function:

$$f_X(x) = \frac{1}{2}e^{-|x|}$$

Now we have to quantize the pdf by using the given thresholds and quantized values with given N and h variables. the **calculateDistortion** function takes N and h as input and returns the value of distortion

first of all it Defines the thresholds and quantized values based on given inputs.

Based on the threshold and quantized values formula, it creates an array of thresholds and also, an array of quantizedValues.

We have to calculate $E[(X - q(X))^2]$ which is defined as below:

$E[(X - q(X))^2] = \text{Sum of integrals over intervals } R_i \text{ of the function } (x - x_i)^2 * \frac{1}{2}e^{-|x|}$
Furthermore, x_i is the quantization value corresponding to the quantization interval R_i in the quantizedValues .

We need to iterate over regions R_i which is obtain by $\text{length}(\text{thresholds}) - 1$, then the integral of equation over intervals including $-\infty$ to $+\infty$ should be calculated, out intervals of R_i exist, we define our equation as below in the Matlab code:

$$\text{func} = \int_{-\infty}^{+\infty} (x - x_i)^2 * \frac{1}{2}e^{-|x|} dx$$

and then, we sum the result of the integral in each iteration which means the range of thresholds(i) and thresholds(i+1) and calculate it as the distortion of given N and h . Furthermore, thresholds values range between $[-\infty \dots R_i \text{ regions} \dots + \infty]$.

For calculating the closed form of the integral, we should partition it. We just calculate the integral from $-\infty$ to 0 and double it, as we have a symmetrical function.

$$\begin{aligned} & \int_{t_{i-1}}^{t_i} (x - x_i)^2 * \frac{1}{2}e^{-|x|} dx \\ &= \int_{t_{i-1}}^{t_i} x^2 * \frac{1}{2}e^x dx + \int_{t_{i-1}}^{t_i} x_i^2 * \frac{1}{2}e^x dx - \int_{t_{i-1}}^{t_i} 2xx_i * \frac{1}{2}e^x dx \\ &= \frac{1}{2} \left(\int_{t_{i-1}}^{t_i} x^2 * e^x dx + x_i^2 * \int_{t_{i-1}}^{t_i} e^x dx - 2x_i * \int_{t_{i-1}}^{t_i} x * e^x dx \right) \\ &= \frac{1}{2} * (e^{t_i} (t_i^2 - 2t_i(q_i + 1) + q_i(q_i + 2) + 2) - e^{t_{i-1}} (t_{i-1}^2 - 2t_{i-1}(q_i + 1) + q_i(q_i + 2) + 2)) \end{aligned} \tag{1}$$

Also when the interval is $(\infty, t1)$ we have:

$$\begin{aligned} & \int_{-\infty}^{t_i} (x - x_i)^2 * \frac{1}{2}e^{-|x|} dx \\ &= \int_{-\infty}^{t_i} x^2 * \frac{1}{2}e^x dx + \int_{-\infty}^{t_i} x_i^2 * \frac{1}{2}e^x dx - \int_{-\infty}^{t_i} 2xx_i * \frac{1}{2}e^x dx \\ &= \frac{1}{2} \left(\int_{-\infty}^{t_i} x^2 * e^x dx + x_i^2 * \int_{-\infty}^{t_i} e^x dx - 2x_i * \int_{-\infty}^{t_i} x * e^x dx \right) \\ &= \frac{1}{2} * e^{t_i} (t_i^2 - 2t_i(q_i + 1) + q_i(q_i + 2) + 2) \end{aligned} \tag{2}$$

Then, for transforming N into R for calculating the rate we use this concept, first of all, we take the number of thresholds N , so this gives us the number of levels which is $N + 1$ levels, then we take $\log_2$ of this value, Then $R(N)$ define as belo

$$R(N) = \log_2(N + 1)$$

Now we can plot the corresponding rate distortion function for the given range of N and different values of h , To do this, we iterate between the given range of $N = [3, 5, 7, \dots, 31]$ and for each value of h , we calculate the distortion and the rate using the function and formulas, and we plot the corresponding rate distortion function as below:

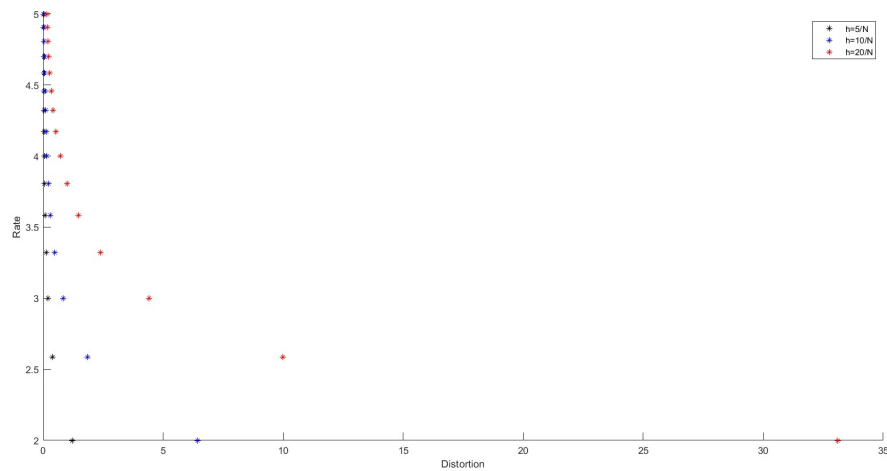


Figure 1: Rate-Distortion function plot of given N , h values

The Matlab code which perform this is as below:

```
1 %% calculate the distortion for each given N,h and then plotting the
2 % results
3 for N = (1:15)*2 + 1
4     c = 0;
5     for h = [5/N, 10/N, 20/N]
6
7         distortion = calculateDistortion(N,h);
8         Rate = log2(N+1);
9         figure(1);
10        hold on
11        if c == 0
12            plot(distortion, Rate, '*k');
13        elseif c == 1
14            plot(distortion, Rate, '*b');
15        else
16            plot(distortion, Rate, '*r');
17        end
18        c = c + 1;
19    end
20
21 end
22 legend({'h=5/N', 'h=10/N', 'h=20/N'})
```

```

23 xlabel('Distortion');
24 ylabel('Rate');
25 hold off;
26
27
28 %% calculates the distortion based on the mentioned integral
29 function distortion = calculateDistortion(N, h)
30     % Define the thresholds and quantized values
31     thresholds = (2*(1:N) - N - 1) * h;
32     thresholds = [-inf thresholds inf];
33     quantizedValues = (2*(1:N+1) - N - 2) * h;
34
35     % Calculate the distortion
36     distortion = 0;
37
38     for i = 1:floor(length(thresholds)/2)
39         qi = quantizedValues(i);
40         if thresholds(i) == -inf
41             ti = thresholds(i);
42             ti1 = thresholds(i+1);
43             temp = 1/2 * ( ...
44                 exp(ti1) * (ti1^2 - 2*ti1*(qi+1) + qi*(qi+2) +2) ...
45                 );
46             distortion = distortion + temp;
47         else
48             ti = thresholds(i);
49             ti1 = thresholds(i+1);
50             temp = 1/2 * ( ...
51                 exp(ti1) * (ti1^2 - 2*ti1*(qi+1) + qi*(qi+2) +2) ...
52                 -exp(ti) * (ti^2 - 2*ti*(qi+1) + qi*(qi+2) +2)...
53                 );
54             distortion = distortion + temp;
55         end
56     end
57     distortion = distortion *2;
58 end

```

Distortion measures the quality of the representation of the data. It quantifies the difference between the original data and its compressed or transmitted version. Lower distortion corresponds to a more accurate or faithful representation of the original data.

we can see the decreasing the distortion values and downward trend in the distortion - rate plots for different h with increasing the N values.

Reducing distortion values typically involves improving the accuracy or fidelity of the representation of information.

It seems that the downward trend of distortion - rate plot when $h = \frac{20}{N}$ is more significant than others and when $h = \frac{5}{N}$ it's so slight. It means that increasing the h value, will make the trend of distortion more quick and also will decline the accuracy of representation of information. As a result, the accuracy of distortion with $h = \frac{5}{N}$ is more than others and it's optimal. specially, between first point and 4th point of N (count from right to left).

The rate refers to the amount of information or the number of bits needed to repre-

sent the data. In the context of compression or communication, it's the rate at which information is transmitted or stored. A higher rate generally implies more detailed or accurate representation.

The rate-distortion trade-off is a fundamental concept in information theory and is often represented by a curve that shows how the distortion changes with variations in the rate. This curve is known as the rate-distortion curve.

The goal in designing compression or communication systems is to find an optimal point on the rate-distortion curve that balances the desire for a low rate (efficient use of resources) and low distortion (high quality of representation). The curve typically exhibits diminishing returns, meaning that achieving lower distortion requires a disproportionately higher rate.

When h is $\frac{20}{N}$, it seems that 4th point is optimal. At this point, distortion has decreased enough (compared to its previous points) and still has a low rate. In the points after that, it is almost less compared to this point, but their rates have increased.

When h is $\frac{10}{N}$, it seems that 2nd point is optimal. At this point, distortion has decreased enough (compared to its previous point) and there isn't significant difference between distortion of 2nd point and its next points. But the advantage of this point is it has lower rate comparing its next points.

When h is $\frac{5}{N}$, it seems that 1st point is optimal. At this point, distortion is low enough and there isn't significant difference between distortion of 1st point and its next points. But the advantage of this point is it has lowest rate comparing its next points.

2 Distortion in two dimensions

Considering the random vector $(X, Y)^T$ with the following probability density function:

$$f_{XY}(x, y) = \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}}$$

Now we want to quantize both horizontal and vertical axes by using the given thresholds and quantized values with given N and h variables using a Matlab function, the **calculateDistortion** function which is provided below takes N and h as input and returns the value of distortion

first of all it Defines the thresholds and quantized values based on given inputs.

Based on the threshold and quantized values formula, it creates an array of thresholds and also, an array of quantizedValues, it is necessary to mention that we append two values of $-\infty$ and $+\infty$ to the thresholds arrays, because we want to apply integral over $-\infty$ to $+\infty$ of the function to calculate the distortion.

It is also contain integral over $-\infty$ to the intervals R_i and to the $+\infty$.

We use same type of quantization for both X and Y two dimensional distribution, we have to calculate $2 * E[(X - q(X))^2]$ which $E[(X - q(X))^2]$ introduce as below:

$$\begin{aligned}
D &= 2 * E[(X - q(X))^2] \\
&= 2 * \sum_{i=1}^N \int_{t_{i-1}}^{t_i} \int_{-\infty}^{\infty} (x - x_i)^2 f_{XY}(x, y) dy dx \\
&= 2 * \sum_{i=1}^N \int_{t_{i-1}}^{t_i} \frac{(x - x_i)^2}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}} dy dx \\
&= 2 * \sum_{i=1}^N \int_{t_{i-1}}^{t_i} (x - x_i)^2 * \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} dx
\end{aligned} \tag{3}$$

In addition, x_i is the quantization value corresponding to the quantization interval R_i in the quantizedValues array which we generate earlier.

The number of quantizedValues is always more than the number of thresholds by 1, so we need to iterate over regions R_i which is obtain by $length(thresholds) - 1$, meanwhile we have to calculate integral of mentioned equation over intervals including $-\infty$ to $+\infty$ and in this range, out intervals of R_i exist, we define our equation as below in the Matlab code:

$$func = 2 \int_{-\infty}^{+\infty} (x - x_i)^2 * \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} dx$$

and then, we sum the result of the integral in each iteration which means the range of thresholds(i) and thresholds(i+1) and calculate it as the distortion of given N and h . Furthermore, thresholds values range between $[-\infty....R_i \text{ regions}.... + \infty]$.

Then, for transforming N into R for calculating the rate we use this concept, first of all, we take the number of thresholds N , so this gives us the number of levels which is $N + 1$ levels, in this exercise we have $N + 1$ horizontal level and $N + 1$ vertical level, so in total we have $(N + 1)^2$ levels, then we take $\log_2$ of this value, Then $R(N)$ define as below:

$$\begin{aligned}
R(N) &= \log_2((N + 1)^2) \\
&= 2 \log_2(N + 1)
\end{aligned}$$

Now we can plot the corresponding rate distortion function for the given range of N and different values of h , To do this, we iterate between the given range of $N = [3, 5, 7, ..., 31]$ and for each value of h , we calculate the distortion and the rate using the function and formulas, and we plot the corresponding rate distortion function as below:

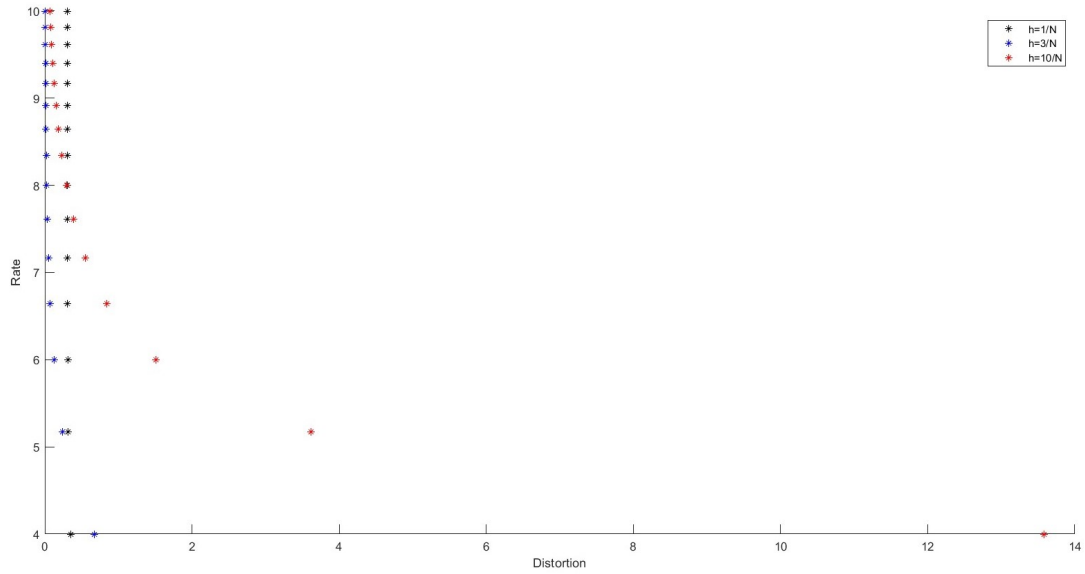


Figure 2: Rate-Distortion function plot of given N , h values

The Matlab code which perform this is as below:

```

1 %% calculate the distortion for each given N,h and then plotting the
2 % results
3 for N = (1:15)*2 + 1
4     c = 0;
5     for h = [1/N, 3/N, 10/N]
6         distortion = calculateDistortion(N,h);
7         Rate = 2*log2(N+1);
8         figure(1);
9         hold on
10        if c == 0
11            plot(distortion, Rate, '*k');
12        elseif c == 1
13            plot(distortion, Rate, '*b');
14        else
15            plot(distortion, Rate, '*r');
16        end
17        c = c + 1;
18    end
19
20 end
21 legend({'h=1/N', 'h=3/N', 'h=10/N'})
22 xlabel('Distortion');
23 ylabel('Rate');
24 hold off;
25
26
27 %% calculates the distortion based on the mentioned integral
28 function distortion = calculateDistortion(N, h)
29     % Define the thresholds and quantized values
30     thresholds = (2*(1:N) - N - 1) * h;
31     thresholds = [-inf thresholds inf];
32     quantizedValues = (2*(1:N+1) - N - 2) * h;
33
34     % Calculate the distortion

```

```

35     distortion = 0;
36
37     for i = 1:length(thresholds)-1
38         qi = quantizedValues(i);
39         func = @(x) 2 * (x-qi).^2 .* exp(-(x.^2)/2)/(sqrt(2.*pi()));
40         distortion = distortion + integral(func, thresholds(i) ...
41             , thresholds(i+1));
42     end
43 end

```

Distortion is a metric of the evaluation the fidelity of data representation, measuring the disparity between the initial data and its transmitted form. Reduced distortion indicates a more precise and faithful rendering of the original data.

The distortion-rate plots illustrate a downward trend as N values increase for various values of h .

Also, Lowering distortion values generally entails enhancing the precision or improvement of information representation.

From the plot we can observe that downward trend of the distortion rate when $h = \frac{10}{N}$ is more remarkable than the two other values of h especially $h = \frac{1}{N}$ trend is very slight.

So it can be obtain that by increasing the amount of h , the trend of distortion will have more Slope and as a consequence the accuracy of representation of information will decrease. As a result, the accuracy of distortion with $h = \frac{1}{N}$ is more than others and it's optimal.

The rate refers to the amount of information or the number of bits needed to represent the data. In the context of compression or communication, a higher rate typically indicates a more detailed or accurate representation.

The rate-distortion is often represented by a curve that shows how the distortion changes with variations in the rate. This curve is known as the rate-distortion curve.

The goal in designing compression or communication systems is to find an optimal point on the rate-distortion curve that balances the desire for a low rate which means the efficient use of resources and low distortion which means high quality of representation.

At $h = \frac{10}{N}$, it seems that 4th (or 5th) point (count from right to left) is optimal. At this point, distortion has decreased enough (compared to its previous points) and still has a low rate. In the points after that, it is almost less compared to this point, but their rates have increased.

At $h = \frac{3}{N}$, it seems that 3rd (or 4th) point (count from right to left) is optimal. At this point, distortion has decreased enough (compared to its previous points) and still has a low rate. In the points after that, it is almost less compared to this point, but their rates have increased.

At $h = \frac{1}{N}$, it seems that 1st (or 2nd) point (count from right to left) is optimal. At this point, distortion has decreased enough (compared to its previous points) and still has a low rate. In the points after that, it is almost less compared to this point, but their rates have increased.

3 Lloyd Algorithm

The code for this task consist of four functions. In the "lloyd" function the Lloyd algorithm steps are implemented and in the output we have the threshold, quantization levels and the average quantization error. The function "calculate_xi" will calculate the

$x_i = \frac{\int_{t_{i-1}}^{t_i} x f_X(x) dx}{\int_{t_{i-1}}^{t_i} f_X(x) dx}$ based on the calculated closed form. For calculating the closed form of x_i we should partition the function as below:

When $t_{i-1} > 0, t_i > 0$ if $t_i \neq \infty$:

$$\frac{\int_{t_{i-1}}^{t_i} x f_X(x) dx}{\int_{t_{i-1}}^{t_i} f_X(x) dx} = \frac{(t_{i-1} + 1) * \exp(-t_{i-1}) - (t_i + 1) * \exp(-t_i)}{\exp(-t_{i-1}) - \exp(-t_i)}$$

, And if $t_i = \infty$ since $\exp(-t_i) = 0$:

$$\frac{\int_{t_{i-1}}^{t_i} x f_X(x) dx}{\int_{t_{i-1}}^{t_i} f_X(x) dx} = \frac{(t_{i-1} + 1) * \exp(-t_{i-1})}{\exp(-t_{i-1})}$$

When $t_{i-1} < 0, t_i > 0$:

$$\frac{\int_{t_{i-1}}^{t_i} x f_X(x) dx}{\int_{t_{i-1}}^{t_i} f_X(x) dx} = \frac{1 - (t_i + 1) * \exp(-t_i) - \exp(t_{i-1}) * (t_{i-1} - 1) - 1}{\sinh(t_i) - \cosh(t_i) + 1 + 1 - \exp(t_{i-1})}$$

, Finally when $t_{i-1} < 0, t_i < 0$ if $t_{i-1} \neq -\infty$ the integral will be the same to it's positive correspondence interval, And if $t_{i-1} = -\infty$ also we can use it's correspondence formula of when $t_i = \infty$.

Also, the function "norm2" simply returns $\|x^{(n+1)} - x^{(n)}\|^2$ that is used as the convergence threshold. Finally the function "calculate_distortion" is used for calculating the average mean squared error with closed form of the formula below:

$$D = \sum_{i=1}^N \int_{t_{i-1}}^{t_i} (x - x_i)^2 f_X(x) dx$$

For calculating the closed form of D, we have the same intuition as calculation of x_i .

When $t_{i-1} > 0, t_i > 0$ if $t_i \neq \infty$:

$$\begin{aligned} \int_{t_{i-1}}^{t_i} (x - x_i)^2 f_X(x) dx &= \frac{1}{2} * \left(\int_{t_{i-1}}^{t_i} x^2 e^{-x} dx + \int_{t_{i-1}}^{t_i} x_i^2 e^{-x} dx - \int_{t_{i-1}}^{t_i} 2xx_i e^{-x} dx \right) \\ &= \frac{1}{2} * \left(\exp(-t_{i-1}) * (t_{i-1}^2 - 2 * (t_{i-1} + 1) * x_i + 2 * t_{i-1} + x_i^2 + 2) \right. \\ &\quad \left. + \exp(-t_i) * (x_i * (2 * t_i - x_i + 2) - t_i * (t_i + 2) - 2) \right) \end{aligned} \quad (4)$$

, And if $t_i = \infty$ since $\exp(-t_i) = 0$:

$$\begin{aligned} \frac{\int_{t_{i-1}}^{t_i} x f_X(x) dx}{\int_{t_{i-1}}^{t_i} f_X(x) dx} &= \int_{t_{i-1}}^{\infty} (x - x_i)^2 f_X(x) dx \\ &= \frac{1}{2} * \left(\int_{t_{i-1}}^{\infty} x^2 e^{-x} dx + \int_{t_{i-1}}^{\infty} x_i^2 e^{-x} dx - \int_{t_{i-1}}^{\infty} 2xx_i e^{-x} dx \right) \\ &= \frac{1}{2} * \exp(-t_{i-1}) * (-2 * (t_{i-1} + 1) * x_i + t_{i-1} * (t_{i-1} + 2) + x_i^2 + 2) \end{aligned} \quad (5)$$

For negative thresholds we simply reverse the threshold and make it positive, and use the formulas above.

Numerical Results:

–For N = 10

◦ Init values :

Thresholds =
 [-Inf, -4.5000, -3.5000, -2.5000, -1.5000, -0.5000, 0.5000, 1.5000, 2.5000, 3.5000, 4.5000, Inf]
 X = -5 -4 -3 -2 -1 0 1 2 3 4 5

◦ Final quantization values :

Thresholds =
 -Inf -4.8966 -3.2519 -2.1115 -1.1710 -0.3680 0.3680 1.1710 2.1115 3.2519 4.8966
 Inf
 X = -5.8966 -3.8584 -2.5756 -1.5686 -0.7164 0 0.7164 1.5686 2.5756 3.8584 5.8966
 Average quantization error = 0.0691

–For N = 20

◦ Init values :

Thresholds =
 -Inf -9.5000 -8.5000 -7.5000 -6.5000 -5.5000 -4.5000 -3.5000 -2.5000 -1.5000 -0.5000
 0.5000 1.5000 2.5000 3.5000 4.5000 5.5000 6.5000 7.5000 8.5000 9.5000 Inf
 X =
 -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8
 9 10

◦ Final quantization values :

Thresholds =
 -Inf -7.4305 -5.7476 -4.6045 -3.6981 -2.9287 -2.2534 -1.6518 -1.1131 -0.6311 -0.2019
 0.2019 0.6311 1.1131 1.6518 2.2534 2.9287 3.6981 4.6045 5.7476 7.4305 Inf
 X =
 -8.4305 -6.3635 -5.0695 -4.0838 -3.2646 -2.5534 -1.9227 -1.3584 -0.8528 -0.4012 0
 0.4012 0.8528 1.3584 1.9227 2.5534 3.2646 4.0838 5.0695 6.3635 8.4305
 Average quantization error = 0.0227

–For N = 100

◦ Init values :

Thresholds =
 -Inf -49.5000 -48.5000 -47.5000 -46.5000 -45.5000 -44.5000 -43.5000 -42.5000 -41.5000
 -40.5000
 -39.5000 -38.5000 -37.5000 -36.5000 -35.5000 -34.5000 -33.5000 -32.5000 -31.5000 -
 30.5000 -29.5000
 -28.5000 -27.5000 -26.5000 -25.5000 -24.5000 -23.5000 -22.5000 -21.5000 -20.5000 -
 19.5000 -18.5000
 -17.5000 -16.5000 -15.5000 -14.5000 -13.5000 -12.5000 -11.5000 -10.5000 -9.5000 -
 8.5000 -7.5000
 -6.5000 -5.5000 -4.5000 -3.5000 -2.5000 -1.5000 -0.5000 0.5000 1.5000 2.5000 3.5000
 4.5000 5.5000 6.5000 7.5000 8.5000 9.5000 10.5000 11.5000 12.5000 13.5000 14.5000

15.5000 16.5000 17.5000 18.5000 19.5000 20.5000 21.5000 22.5000 23.5000 24.5000
 25.5000
 26.5000 27.5000 28.5000 29.5000 30.5000 31.5000 32.5000 33.5000 34.5000 35.5000
 36.5000
 37.5000 38.5000 39.5000 40.5000 41.5000 42.5000 43.5000 44.5000 45.5000 46.5000
 47.5000
 48.5000 49.5000 Inf
 X =
 -50 -49 -48 -47 -46 -45 -44 -43 -42 -41 -40 -39 -38 -37 -36 -35 -34 -33 -32
 -31 -30 -29 -28 -27 -26 -25 -24 -23 -22 -21 -20 -19 -18 -17 -16 -15 -14 -13
 -12 -11 -10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6
 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44
 45 46 47 48 49 50

◦ **Final quantization values :**

Thresholds =
 -Inf -18.8132 -17.1706 -16.0803 -15.2314 -14.5154 -13.8818 -13.3029 -12.7627 -12.2510
 -11.7614
 -11.2895 -10.8326 -10.3886 -9.9562 -9.5344 -9.1228 -8.7207 -8.3279 -7.9441 -7.5692 -
 7.2030
 -6.8454 -6.4964 -6.1558 -5.8236 -5.4998 -5.1843 -4.8771 -4.5780 -4.2871 -4.0042 -3.7294
 -3.4625 -3.2035 -2.9523 -2.7089 -2.4731 -2.2450 -2.0244 -1.8113 -1.6056 -1.4071 -1.2159
 -1.0317 -0.8545 -0.6843 -0.5209 -0.3641 -0.2139 -0.0702 0.0702 0.2139 0.3641 0.5209
 0.6843 0.8545 1.0317 1.2159 1.4071 1.6056 1.8113 2.0244 2.2450 2.4731 2.7089
 2.9523 3.2035 3.4625 3.7294 4.0042 4.2871 4.5780 4.8771 5.1843 5.4998 5.8236
 6.1558 6.4964 6.8454 7.2030 7.5692 7.9441 8.3279 8.7207 9.1228 9.5344 9.9562
 10.3886 10.8326 11.2895 11.7614 12.2510 12.7627 13.3029 13.8818 14.5154 15.2314
 16.0803
 17.1706 18.8132 Inf
 X =
 -19.8132 -17.7765 -16.5283 -15.5965 -14.8310 -14.1654 -13.5646 -13.0086 -12.4851 -
 11.9863 -11.5070
 -11.0437 -10.5942 -10.1568 -9.7305 -9.3145 -8.9083 -8.5115 -8.1237 -7.7449 -7.3749 -
 7.0135
 -6.6607 -6.3164 -5.9805 -5.6530 -5.3338 -5.0229 -4.7201 -4.4255 -4.1390 -3.8605 -3.5900
 -3.3274 -3.0726 -2.8256 -2.5864 -2.3547 -2.1307 -1.9141 -1.7049 -1.5031 -1.3084 -1.1210
 -0.9405 -0.7670 -0.6004 -0.4404 -0.2872 -0.1404 0 0.1404 0.2872 0.4404 0.6004
 0.7670 0.9405 1.1210 1.3084 1.5031 1.7049 1.9141 2.1307 2.3547 2.5864 2.8256
 3.0726 3.3274 3.5900 3.8605 4.1390 4.4255 4.7201 5.0229 5.3338 5.6530 5.9805
 6.3164 6.6607 7.0135 7.3749 7.7449 8.1237 8.5115 8.9083 9.3145 9.7305 10.1568
 10.5942 11.0437 11.5070 11.9863 12.4851 13.0086 13.5646 14.1654 14.8310 15.5965
 16.5283
 17.7765 19.8132
 Average quantization error = 0.0027

```
1 N = 100;
2 digits(50000);
3 [X, thresholds] = lloyd(N);
4 distortion = calculate_distortion(X, thresholds);
```

```

5
6 %% implementation of lloyd algorithm
7 function [X, thresholds] = lloyd(N)
8     % initialize the thresholds and quantized values as mentioned in
9     % the question
10    thresholds = -(N-1)/2:1:(N-1)/2;
11    thresholds = [-inf thresholds inf];
12    X = -N/2:1:N/2;
13
14    disp("Init values :");
15    disp("Thresholds = ");
16    disp(thresholds);
17    disp("X = ")
18    disp(X);
19    i = 1;
20    while true
21        distortion = calculate_distortion(X, thresholds);
22        % disp(distortion);
23        Xprev = X;
24        % Calculating new thresholds
25        for j = 1:N
26            thresholds(j+1) = (X(j) + X(j+1)) / 2;
27        end
28
29        % Calculating new X
30        for j = 1:N+1
31            X(j) = calculate_xi(thresholds(j+1), thresholds(j-1+1));
32        end
33        % Checking the end statement
34        i = i + 1;
35        if norm2(X, Xprev) < 0.01
36            break;
37        end
38    end
39    disp("Final quantization values :");
40    disp("Thresholds = ");
41    disp(thresholds);
42    disp("X = ")
43    disp(X);
44    disp("Average quantization error = ");
45    disp(distortion);
46 end
47
48
49 %% calculating ||x^(n+1)-x^(n)||^2
50 function output = norm2(x, y)
51     output = sum((x-y).^2);
52 end
53
54 %% calculate xi using the closed form of integral
55 % more details of the closed form is written in the report
56 function xi = calculate_xi(ti, ti_1)
57     if ti > 0 && ti_1 > 0
58         if ti == inf
59             integral_xfx = (ti_1 + 1)*exp(-ti_1);
60             integral_fx = exp(-ti_1);
61         else
62             integral_xfx = (ti_1 + 1)*exp(-ti_1) - (ti + 1)*exp(-ti);

```

```

63         integral_fx = exp(-ti_1) - exp(-ti);
64     end
65 elseif ti > 0 && ti_1 < 0
66     integral_xfx = 1 - (ti + 1)*exp(-ti) - exp(ti_1)*(ti_1 - 1) -
67 1;
68     integral_fx = sinh(ti) - cosh(ti) + 1 + 1 - exp(ti_1);
69 else
70     if ti_1 == -inf
71         integral_xfx = (ti - 1)*exp(ti);
72         integral_fx = exp(ti);
73     else
74         integral_xfx = (ti - 1)*exp(ti) - (ti_1 - 1)*exp(ti_1);
75         integral_fx = exp(ti) - exp(ti_1);
76     end
77     xi = integral_xfx / integral_fx;
78 end
79
80 %% calculating the distortion using the closed form of D
81 % more details of the closed form is written in the report
82 function d = calculate_distortion(X, thresholds)
83     d = 0;
84     for i=1:length(thresholds)-1
85
86         if thresholds(i) < 0 && thresholds(i+1) < 0
87             ti_1 = -thresholds(i+1);
88             ti = -thresholds(i);
89             xi = -X(i);
90             if ti ~= inf
91                 d = d + 1/2*(exp(-ti_1) * ( ti_1^2 - 2*(ti_1+1)*xi ...
92                     + 2*ti_1 + xi^2 + 2 ) ...
93                     + exp(-ti) * ( xi*(2*ti - xi + 2) - ti*(ti+2) - 2 )
94             );
95             else
96                 d = d + 1/2*exp(-ti_1)*( -2*(ti_1+1)*xi + ti_1*(ti_1+2)
97                 ...
98                     + xi^2 + 2);
99             end
100         elseif thresholds(i) < 0 && thresholds(i+1) > 0
101             ti = thresholds(i+1);
102             d = d + 2*(1- 1/2*(ti*(ti+2)+2)*exp(-ti));
103         else
104             ti_1 = thresholds(i);
105             ti = thresholds(i+1);
106             xi = X(i);
107             if ti ~= inf
108                 d = d + 1/2*(exp(-ti_1) * ( ti_1^2 - 2*(ti_1+1)*xi + 2*
109 ti_1 ...
110                     + xi^2 + 2 ) ...
111                     + exp(-ti) * ( xi*(2*ti - xi + 2) - ti*(ti+2) - 2 )
112             );
113             else
114                 d = d + 1/2*exp(-ti_1)*( -2*(ti_1+1)*xi + ti_1*(ti_1+2)
115                 ...
116                     + xi^2 + 2);
117             end
118         end
119     end
120 end

```

